

Cost function (Hall-sensor-based, after eliminating d):

$$\text{cost} \Rightarrow J(\hat{\ell}) = \sum_{j=1}^{N_I} \left[\sum_{i=1}^{N_p} \mathbf{b}_{ij}^T \mathbf{b}_{ij} - \left(\sum_{i=1}^{N_p} \mathbf{v}_j^T \mathbf{S}_i^T \mathbf{b}_{ij} \right)^T \left(\sum_{i=1}^{N_p} \mathbf{v}_j^T \mathbf{S}_i^T \mathbf{S}_i \mathbf{v}_j \right)^{-1} \left(\sum_{i=1}^{N_p} \mathbf{v}_j^T \mathbf{S}_i^T \mathbf{b}_{ij} \right) \right]$$